

712.

A PARTIAL DIFFERENTIAL EQUATION CONNECTED
WITH THE
SIMPLEST CASE OF ABEL'S THEOREM.

[From the *Report of the British Association for the Advancement of
Science*, (1881), pp. 534, 535.]

CONSIDER a given cubic curve cut by a line in the points (x, p) , (x_1, p_1) , (x_2, p_2) ; using the first and second points as elements, these determine uniquely the third point. Analytically, the equation of the curve determines p as a function of x , and p_2 as a function of x_1 ; writing in the equation

$$c_2 = \lambda c_1 + (1 - \lambda)c, \quad y_2 = \lambda y_1 + (1 - \lambda)y_0,$$

we have c , by a simple equation, and thence p_2 ; viz. x_2 is found as a function of x , x_1 , and of the nine constants of the equation. Hence having the derived equations (in regard to x , x_1) of the first, second, and third orders, we have $(1 + 2 + 3 + 4) = 10$ equations from which to eliminate the 6 quantities x_2 , considered as a function of x and x_1 , thus involves a partial differential equation of the third order, independent of the particular cubic curve.

To obtain this equation it is only necessary to observe that we have, by Abel's theorem,

$$\frac{dx}{X} + \frac{dx_1}{X_1} + \frac{dx_2}{X_2} = 0,$$

where X , X_1 , X_2 is a given function of x , and p , that is, of x ; X_1 and X_2 are the like functions of x_1 and x_2 respectively. Hence, considering x_2 as a function of x and x_1 , we have

$$\frac{dx_2}{dx} = -\frac{X_2}{X}, \quad \frac{dx_2}{dx_1} = -\frac{X_2}{X_1},$$

[C. XI. 4-2]

and consequently

$$\frac{dx_2}{dx_1} \div \frac{dx_2}{dx} = \frac{X}{X_1},$$

where X , X_1 are functions of x_2 , x_1 respectively; hence taking the logarithm and differentiating successively with regard to x_1 and x_2 , we have

$$\frac{d}{dx_2} \frac{d}{dx_1} \log \left(\frac{dx_2}{dx_1} \div \frac{dx_2}{dx} \right) = 0,$$

which is the required partial differential equation of the third order.

This differential equation has a simple geometrical significance. Consider three consecutive positions of the line meeting the cubic curve in the points 1, 2, 3; 1', 2', 3'; 1'', 2'', 3'' respectively; *quæ* equation of the third order, the equation should in effect determine 3'' by means of the other points. And, in fact, the three positions of the line constitute a cubic curve; the nine points are thus the intersections of two cubic curves, or, say, they are an “annexed” of points; any eight of the points thus determine uniquely the ninth point.

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713.

ADDITION TO MR ROWE'S MEMOIR ON ABEL'S THEOREM.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII., Part III. (1881), pp. 715—758. Received May 27,—Read June 16, 1880.]

IN Abel's general theorem y is an irrational function of x determined by an equation $\chi(y) = 0$, or say $\chi(x, y) = 0$; if the order n as regards y , and that is also by him that the sum of any number of the integrals considered may be reduced to a sum of ρ integrals; where ρ is a determinate number depending only on the form of

the equation $\chi(x, y) = 0$, and given in his equation (42), [(*Euvres Complètes*, (1881), t. I., p. 157]; viz. if, solving the equation so as to obtain from it developments of y in descending series of powers of x , we have:

$$\begin{array}{ll} v_{\sigma}\rho_{\nu}, & \text{series each of the form } y = x^{\sigma'_{\nu}} + \dots, \\ & v_{\sigma}\rho_2, \quad y = x^{\sigma'_2} + \dots, \\ & \vdots \quad \quad \quad \vdots \\ & v_{\sigma}\rho_n, \quad y = x^{\sigma'_n} + \dots, \end{array}$$

⁰The second power of a term coefficient: this being a relation $y = x^p + \dots$, representing the y , different values of y obtained by giving to the radical x^p and to its p values, and the corresponding values in the radicals which enter into representing y , values of y . It is assumed that the equation is such that the meaning is to make each representing y , y_2 . In each of the equations the meaning of y is taken positive in the development of y in descending powers of x .

(so that $\sigma = v_1\rho_1 + v_2\rho_2 + \cdots + v_n\rho_n$) then y is a determinate function of $x_1, x_2, x_3, x_4, x_5, x_6, \dots, x_n$, say ρ .

We have so expressed Abel's ρ in the following form, viz. assuming

$$y = \Sigma a_1 \alpha_1 \nu_1 + \Sigma a_2 \nu_2 \cdots + \Sigma a_n \nu_n - \frac{1}{2} \Sigma a - \frac{1}{2} n + 1,$$

or, what is the same thing, for ρ writing this Σa_1 ,

$$y = \Sigma a_1 \nu_1 + \Sigma a_2 \nu_2 + \Sigma a_3 \nu_3 + \cdots + \Sigma a_n \nu_n - \frac{1}{2} \Sigma a - \frac{1}{2} n + 1,$$

where in the first sum ν_1 have each of them the values $1, 2, \dots, \rho_1$ subject to the condition $i + \nu_1 \sigma'_1$; in each of the other sums ν_i and ρ_n are interrelated by the suffix, which has the values $1, 2$, etc.

It is a leading result in Riemann's theory of the Abelian integrals that y is the deficiency (Geschlecht) of the curve represented by the equation $\chi(x, y) = 0$; and it must consequently be demonstrable *a posteriori* that the foregoing expression for y is in fact = deficiency of curve $\chi(x, y) = 0$. I propose to verify this by means of the formulae given in my paper "On the Higher Singularities of a Plane Curve," *Quart. Math. Journ.*, vol. VII., (1866), pp. 212—222, [374].

It is necessary to distinguish between the values of $\frac{m}{n}$ which are $>$, $=$, and < 1 ; and to fix the ideas I assume $l = 7$, and

$$\begin{aligned} & \frac{m_1}{n_1}, \frac{m_2}{n_2}, \frac{m_3}{n_3}, \text{ each } > 1, \\ & \frac{m_4}{n_4} = 1; \text{ say } m_4 = n_4 = n, \text{ and } n_4 = n; \\ & \frac{m_5}{n_5}, \frac{m_6}{n_6}, \frac{m_7}{n_7}, \text{ each } < 1, \end{aligned}$$

but it will be easily seen that the reasoning is quite general. And now Σ to denote a sum in regard to the first set of suffixes $1, 2, 3$, and Σ' to denote a sum in regard to the second set of suffixes $5, 6, 7$. The foregoing value of ρ is thus

$$y = \Sigma a \nu + 4 + \Sigma' a \nu.$$

Introducing a third coordinate z for homogeneity, the equation $\chi(x, y) = 0$ of the curve will be

$$0 = [y^{p-(n-1)}, y^{n-2}x^a, \dots] (y, z)^{n-2} + (y^{n-2}, y^{n-3}x^a, \dots) (y, z)^{n-1} + \cdots + (y)$$

where it is to be observed that $\chi(x, y) = 0$ is the product of ν_i , different ovaries; these divide themselves into n_i

groups, each a product of ρ_i orates; and in each such product the ρ_i coefficients A_i are in general the ρ_i values of a function containing a radical a^{1/ρ_i} and are thus different from each other; it is to when follows in effect assumed in each step that all the a_i , but that all the x_i coefficients, etc. are different from each other.

The remarks apply to the other factors. It applies in particular to the factor $(y, z)^{r-1}$, viz. it is assumed that the coefficients A in the radical $z, y^{n-2}, \dots$ are all of them different from each other. These assumptions as to the leading coefficients really imply Abel's assumption that $\frac{m_1}{n_1}, \frac{m_2}{n_2}$ are all of them fractions in their least terms, viz. that $n = 1$, since

1 rotate however for convenience the equation will be considered, by giving to

In the product of the several infinite series, the terms containing the lowest powers of $z, \frac{1}{n_1}$ is a function in its lowest terms, viz. since $n_1 = 1$.

A curve has singularities (singular points) at infinity, that is, on the line $z = 0$; viz. *First*, a singularity at $(x = 0, x' = 0)$, where the tangent is $x' = 0$, and which, writing for convenience $x = 1$, is denoted by the function

$$(z - \nu^{m/n} x^{1-1/n})^{n-1} = \dots$$

where observe that the expressed factor indicates r_n branches $(z - \nu^{m/n} x^{1-1/n})^{r_n}$, or say $r_n(m_n - p_n)$ partial branches $z = \nu^{m/n} \dots$, that is, $r_n(m_n - p_n)$ partial branches $z = \nu^{m/n} \dots$, will in all $r_n(m_n - p_n) + \dots$, where $r_n(m_n - p_n)$ partial branches at the encompassed factors with the suffixes 1' and 4; that is, we can regard the encompassed factors with the suffixes 1' and 4, where in the equation

$$z - \nu^{m/n} + \dots = 0.$$

Secondly, a singularity at $(z = 0, y = 0)$, where the tangent is $y = 0$, and which, writing for convenience $z = 1$, is denoted by the function

$$(y - z^{m/n} x^{1-1/n})^{n-1} = \dots$$

^{0*} This assumption is virtually made by Abel, (1), [c. p. 157], in considers

where observe that the expressed factor indicates r_5 branches $(y - z^{m/n}x^{1-1/n})^{r_5}$, or say $r_5(n_5 - m_5)$ partial branches $y = z^{m/n} \dots$, that is, $r_5(n_5 - m_5)$ partial branches $y = z^{m/n} \dots$, with in all $r_5n_5 + \dots$, with $r_5n_5 + r_6n_6 + r_7n_7$ partial branches $y = z^{m/n} \dots$, with the suffixes 5, 6, 7.

Thirdly, suppose at $(x = 0, y = 0)$, where the tangent is $z = 0$, and consequently $r_4n_4 = 0$, partial branches $z = y^0 \dots +$ similar terms with respect to x, y, z .

It is convenient to assume $r_4 = 2$, so that

$$\begin{aligned} E &= \frac{1}{2}(K - 1)(K - 2) = z, \\ E' &= \frac{1}{2}(K' - 1)(K' - 2) \\ &= \frac{1}{2}\Sigma(n - 1). \end{aligned}$$

hence

$$E + e' = \frac{1}{2}(K - 1) + \frac{1}{2}\Sigma(n - 1).$$

Assuming that there are no other singularities, the deficiency

$$\rho = \frac{1}{2}(K - 1)(K - 2) - E - e'$$

is

$$= \frac{1}{2}(K - 1)(K - 2) - \frac{1}{2}KK + \frac{1}{2}\Sigma(n - 1).$$

This should be equal to the before-mentioned value of ρ ; viz. we ought to have

$$\frac{1}{2}(K - 1)(K - 2) - \frac{1}{2}KK + \frac{1}{2}\Sigma(n - 1) = \Sigma a\nu_1 + 4 + \Sigma' a\nu_2 - \Sigma a - \Sigma' a - 2,$$

or, as it will be convenient to write it,

$$M = K - \frac{3}{2}K + \frac{1}{2}\Sigma(n - 1) - 1 = \Sigma a\nu_1 + \Sigma' a\nu_2 - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of N and $\Sigma(n - 1)$ calculated, according to the method of my paper, for the foregoing singularities of the curve.

the equation $z^n = y^n \sqrt[n]{f(x)}$ as denoting integers; but where it appears that the indices a, b , are to be considered of fixed, and as in the values of degrees with respect to x . He says that if these be assumed as integers in the expression of y in descending powers of x (the leading coefficient being z in regard to y), the singular factor are in (x, y) in regard to y in regard to x .

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The terms $\Sigma n + \Sigma' n + \theta n$, written at length, is

$$\begin{aligned} &= n_1(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) + \cdots + n_2(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_3(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_4(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_5(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_6(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) \\ &+ \cdots + n_7(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3). \end{aligned}$$

which is

$$= \Sigma \nu_1 n_2 + \Sigma' n_2 (2\rho_2 + 2'\rho_1) + \Sigma \Sigma' + \Sigma' \Sigma + \Sigma a_1 a_2 a_3,$$

We have moreover

$$\Sigma' \rho_1 = \Sigma a_1 \rho_1 + (\theta + \Sigma' a_1) \rho_1 = \Sigma a_1 + \Sigma a_2 + \Sigma' a_1 \rho_1,$$

each branch $(z - y^{m/n} x^{1-1/n})^{n-1}$ gives $a = m_1 \rho_1$, and the value of $\Sigma(\kappa - 1)$ for this singularity is

$$n_1(m_1 - n_1 - 1) + r_1 n_1(m_1 - 1) = r_1 n_1 \rho_1 - n_1 \rho_1.$$

which is

$$= \Sigma a \nu_1 - \Sigma a \rho_1.$$

each branch $(y - z^{m/n} x^{1-1/n})^{n-1}$ gives $\sigma = m_n \rho_n$, and the value of $\Sigma(\kappa - 1)$ for this singularity is

$$n_4(m_4 - n_4 - 1) + r_4 n_4(m_4 - 1) = r_4 n_4 \rho_4 - n_4 \rho_4,$$

which is

$$= \Sigma' \nu_2 - \Sigma' \nu_4 - \Sigma' \nu_1.$$

For each of the θ singularities

$$(y - z^{m/n} x^{1-1/n})^{n-1}.$$

we have $a = n$, and the value of $\Sigma(\kappa - 1)$ is $r = \theta(\kappa - 1)$; this is $= 0$ for the value $\lambda = 1$, which is physically attributed to λ .

According to the theory explained in my paper above referred to, the several singularities not together equivalent to a certain number K of κ^* nodes and cusps; viz. we have

$$K = \Sigma n - \Sigma a + \frac{1}{2} \Sigma(\kappa - 1).$$

This should be equal to the before-mentioned value of ρ ; viz. we ought to have

$$(K-1)(K-2) - 2(K + \Sigma n + \Sigma' n + \theta n) = 2\Sigma a \nu_1 + 4 + 2\Sigma a \nu_2 - 2\Sigma a - 2\Sigma' a - 4,$$

or to it will be convenient to write

$$M - K - \Sigma K + 1 + \Sigma(n - 1) = \Sigma a \nu_1 + \Sigma' a \nu_2 - \Sigma a - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of N and $\Sigma(n - 1)$ calculated, according to the method of my paper, for the foregoing singularities of the curve.

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The term $\Sigma n + \Sigma' n + \theta n$, written at length, is

$$= \Sigma \nu_1 n_1 + \Sigma' \nu_2 n_2 + \theta n,$$

[C. XI. 5]

or, reducing,

$$M = (\Sigma m)^2 - \Sigma ma - \Sigma m\rho_1 - \Sigma m\rho_2 + (\Sigma' m)^2 - \Sigma' m_2 a - \Sigma' m_2 \rho_2 \\ + (\Sigma' m)^2 - \Sigma' m\rho_1 - \Sigma ma + 2\Sigma' m_2 a_1 + 2\Sigma' \Sigma a_1 a_2;$$

and it is to be shown that the two lines of this expression are in fact the values of M belonging to the singularities

$$\left(z - y^{m/n} x^{1-1/n}\right)^{n-1}, \dots, \text{ and } \left(y - z^{m/n} x^{1-1/n}\right)^{n-1}, \dots,$$

respectively. We assume $\lambda = 1$, and there is then no singularity $(y - z)^{n-1}$.

I recall that, considering the several partial branches which meet in a singular point, M denotes the sum of the number of the intersections of each partial branch by every other partial branch: so that for each pair of partial branches the intersections are to be counted twice. Supposing that the tangent is $x = 0$, and that, for any two branches we have $z = Az^x + \dots$ (x where p_x, p_x are each up to greater than 1), then if $p_x = p_x$, and $A_x = A_x, A_x = A_x$ where A_x, A_x are not the partial branches which have here always could be regards the cases about to be considered), then the number of intersections is taken to be $-$; and if $p_x > p_x$, and the number of intersections (taken to be $\equiv p_x$) viz. in the case of as a unequal exponents, it is equal to the smaller exponents.

Consider now the singularity $\left(z - y^{m/n} x^{1-1/n}\right)^{n-1}$, and first the intersections of a partial branch $z = y^{m/n} \dots$ by each of the remaining $n_1(m_1 - p_1) - 1$ partial branches of the same set: the number of intersections with any one of these is $= \frac{m_1}{n_1} [n_1(m_1 - p_1) - 1]$. But we obtain this same number from each of the $n_1(m_1 - p_1)$ partial branches, and thus the whole number is

$$n_1(m_1 - p_1) \frac{m_1}{n_1} [n_1(m_1 - p_1) - 1] = n_1 m_1 [n_1(m_1 - p_1) - 1].$$

Taking account of the other sets, and the whole number of intersections is

$$= \Sigma n^2 a - \Sigma' n a^2.$$

Observe now that $\frac{m_1}{n_1} > \frac{m_2}{n_2}$, that is, $\frac{n_1}{n_2} < \frac{m_1}{m_2}$, and that, these being each < 1 , we have $1 - \frac{m_2}{n_2} > 1 - \frac{m_1}{n_1}$, that is, $\frac{n_2 - m_2}{n_2} > \frac{n_1 - m_1}{n_1}$, and we thus have

$$\frac{m_1}{n_1} < \frac{m_2}{n_2}, \text{ and } \left(z - y^{m/n} x^{1-1/n} \right)^{n_1(m_1-p_1)},$$

Considering now the intersections of partial branches of the two sets

$$\left(z - y^{m/n} x^{1-1/n} \right)^{n_1(m_1-p_1)} \text{ and } \left(z - y^{m/n} x^{1-1/n} \right)^{n_2(m_2-p_2)},$$

respectively, a partial branch $z - z^{m/n} \dots$ gives with each partial branch of the other set a number $= \frac{m_2}{n_2}$; and in this way taking each partial branch of each set, the number is

$$n_1(m_1 - p_1) \cdot n_2(m_2 - p_2) \cdot \frac{m_2}{n_2} = n_2 m_2 n_1(m_1 - p_1),$$

and thus for all the sets the number is

$$= 2 \Sigma n_2 m_2 \nu_2 m_2,$$

where in the first sum the Σ' refers to each pair of values of the suffixes. But the intersections are to be taken twice; the number thus is

$$= 2 \Sigma m_1 m_2 \nu_1 \nu_2,$$

Adding the foregoing number

$$= \Sigma' n^2 a - \Sigma' n a^2;$$

the whole number for the singularity in question is

$$= (\Sigma m_1)^2 - \Sigma' m_2 \rho_2 - \Sigma' m_2 \rho_2.$$

Similarly for the singularity $\left(y - z^{m/n} x^{1-1/n} \right)^{n-1} \dots$, taking each set with itself, the number of intersections is

$$n_5 m_5 [n_5(m_5 - p_5) - 1] + n_6 m_6 [n_6(m_6 - p_6) - 1] + n_7 m_7 [n_7(m_7 - p_7) - 1],$$

which is

$$= \Sigma' m_2^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a.$$

$$[\hspace{10em} 5 - 2]$$

We have here $\frac{m_5}{n_5} > \frac{m_6}{n_6}$, each of these being < 1 , we have $1 - \frac{m_5}{n_5} < 1 - \frac{m_6}{n_6}$, that is, $\frac{n_5 - m_5}{n_5} < \frac{n_6 - m_6}{n_6}$, and so

$$\frac{m_5}{n_5} > \frac{m_6}{n_6}, \frac{n_5 - m_5}{n_5} < \frac{n_6 - m_6}{n_6};$$

Hence considering the two sets

$$\left(y - z^{m/n} x^{1-1/n}\right)^{n_5(m_5-p_5)} \quad \text{and} \quad \left(y - z^{m/n} x^{1-1/n}\right)^{n_6(m_6-p_6)};$$

a partial branch of the first set given with a partial branch of the second set a number of intersections; and the number then obtained is

$$n_5(m_5 - p_5) \cdot n_6(m_6 - p_6) \cdot \frac{m_6}{n_6} = n_6 m_6 n_5 (m_5 - p_5),$$

For all the sets the number is

$$n_5 m_5 [n_5(m_5 - p_5) + n_6(m_6 - p_6)] - n_5 n_6 (m_6 - m_5)$$

or taking these from, the number is

$$2\Sigma' m_2 m_2 \nu_2 \nu_2.$$

where in the first sum the Σ' refers to each pair of suffixes. Adding the foregoing value

$$\Sigma' m^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a,$$

the whole number for the singularity in question is

$$= (\Sigma' m_2)^2 - \Sigma m_1 a - \Sigma' m_2 \rho_2.$$

and the proof is thus completed.

Referring to the first-note (ante, p. 31), I remark that the theorem “ ρ -deficiency, is absolute, and applies to a curve with any singularities whatever: in a curve which has singularities not taken account of in Abel’s theory, the “perhaps see particulars” which we have a give as he expresses it, taken account of, viz. for that Σa is, in the value of ρ as denoted by Abel’s formula; and the actual deficiency will be = Abel’s ρ , = such diminution, that is, it will be = true values of ρ .

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714.

VARIOUS NOTES.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 69, 115, 124, 125.]

AN IDENTITY.

THE following remarkable identity is given under a slightly different form by Gauss, *Werke*, t. III., p. 419.

$$\begin{aligned} 1 &= \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \dots \\ &= \left[1 + \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \dots\right]^2. \end{aligned}$$

ON TWO RELATED QUADRIC FUNCTIONS.

Assume

$$\begin{aligned} \varphi x &= x^2(1-x) + x(1-x^2-\alpha x), \\ \varphi y &= y(1-x) + y(x^2-1-\alpha x). \end{aligned}$$

then

$$\begin{aligned} \varphi\left(\frac{x^2}{x-a}\right) &= \frac{x^2}{(x-a)^2} \varphi x, \\ \varphi\left(\frac{x^2}{x-a}\right) &= \frac{x^2}{(x-a)^2} \varphi y. \end{aligned}$$

In the like of those for a ratio $\frac{p}{1-p}$, then

$$\varphi\left[\frac{ax^2(x-y)}{x^2-y^2(x-a)}\right] = \frac{ax^2(x-y)^2}{(x-a)^2-y^2(x-a)^2} \varphi(x).$$

A TRIGONOMETRICAL IDENTITY.

$$\begin{aligned}
 & \cos(b-c) \sin(b+c-a) + \cos a \sin(b+c-a) \\
 & + \cos(c-a) \sin(c+a+b) + \cos b \sin(c+a+b) \\
 & + \cos(a-b) \sin(a+b+c) + \cos c \sin(a+b+c) \\
 = & \cos a \cos b \cos(b+c) + \cos b \cos c \cos(c+a) + \cos c \cos a \cos(a+b) + \cos a \cos b \cos(b+c)
 \end{aligned}$$

EXTRACT FROM A LETTER.

“I wish to construct a correspondence such as

$$(x + iy)^2(x + iy) = X + iY,$$

or, say, for greater convenience

$$4(x + iy)^2(x + iy) = X + iY,$$

viz. if

$$x + iy = \cos \alpha,$$

then

$$X + iY = \cos \alpha \beta.$$

Suppose α , in a value of α corresponding to a given value of $X + iY$, then the three values of $x + iy$ are of course $\cos \alpha$, $\cos(\alpha + \frac{2\pi}{3})$, but I am afraid that the calculation of α , even with red and card tables, would be very laborious. Writing

$$X + iY = R(\cos \Theta + i \sin \Theta),$$

the intervals for Θ might be 3° , 30° or even 15° ; those of R , say 0.1 from 0 to 2, and then 0.2 up to 4; and 2 places of decimals would be quite sufficient; but even this would probably involve a great mass of calculation.

It has occurred to me that perhaps a geometrical solution might be found for the equation $X + iY = \cos \beta$.”

October 21, 1877.

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715.

NOTE ON A SYSTEM OF ALGEBRAICAL
EQUATIONS.[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 17, 18.]

ASSUME

$$\begin{aligned}x + y + z &= P, \\yz + zx + xy &= Q, \\xyz &= R.\end{aligned}$$

$$\begin{aligned}A &= (nx + y - z) - a^n(nx + P), \\B &= y(ax + Q) - b^n(my + Q), \\C &= z(ax + Q) - c^n(nz + P), \\\Theta &= -mnQ + nP.\end{aligned}$$

Then

$$\begin{aligned}(ax + P)B - (my + P)A &= (ax + P)y(ax + Q) - a^n(my + Q) \\&= (ax + P)(ax + Q) - (my + Q)(my + P) \\&= ax(my + Q - ay) - yz(my + Q - P) + yz(ax + P) \\&= mny(ax - y) - P(ax - y) \\&= (ax - y)(mny - P) = (ax - y)\Theta,\end{aligned}$$

whence, identically,

$$\begin{aligned}(ax + P)B - (my + P)A &= (ay - y)\Theta, \\(my + P)C - (nx + P)B &= (by - z)\Theta, \\(ax + P)A - (nx + P)C &= (cz - x)\Theta.\end{aligned}$$

Hence any two of the equations $A = 0$, $B = 0$, $C = 0$ imply the third equation.

We have

$$\begin{aligned} A &= x \left[(n + 1)x + y + z \right] - a^n \left[(n + 1)x + y + z \right] \\ &= (x - a^n) \left[(n + 1)x + y + z \right]; \end{aligned}$$

and similarly for B and C . The three equations therefore are

$$\begin{aligned} \frac{x}{x^n - a^n} &= \frac{y + z}{(n + 1)x^n + (n - 1)a^n}, \\ \frac{y}{y^n - b^n} &= \frac{x + z}{(n + 1)y^n + (n - 1)b^n}, \\ \frac{z}{z^n - c^n} &= \frac{x + y}{(n + 1)z^n + (n - 1)c^n}, \end{aligned}$$

and any two of these equations imply the third equation.

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716.

AN ILLUSTRATION OF THE THEORY OF THE
 ϑ -FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 27—32.]

IF X be a given quartic function of x , and if a , as for convenience a constant multiple as, be the value of the integral $\int \frac{dx}{\sqrt{X}}$, taken from a given inferior limit to the superior limit x ; then, conversely, x is expressible as a function of a , viz. it is expressible in terms of ϑ -functions of a , where ϑa , or say $\vartheta(a, \Omega)$, is the definition of which, depending, depend in general by definition on the sum of a series of exponentials of a , and it is possible from the assumed equation $a = \int \frac{dx}{\sqrt{X}}$, and the definition of ϑa , to attain by general theory the actual formulae for the determination of a as such a function of a .

I propose here to obtain these formula, in the case where X is a product of real factors, in a less suitable manner, viz. by reducing the integral $\int \frac{dx}{\sqrt{X}}$ by a linear substitution in the form of an elliptic integral, the object being to obtain for the same in question the actual formulae as a function of ϑ -functions of u .

The definition of ϑu , when the parameter is expressed, $\vartheta(u, \Omega)$ is

$$\vartheta u = 2\Sigma(-)^p q^{p+\frac{1}{2}(p+1)^2} \sin(p + \frac{1}{2})u,$$

where u and p are negative integer values, zero included, from $-\infty$ to $+\infty$ (that is, from -2 to 2 , 0 to 2 , etc.); the parameter Ω , or Ω' (of imaginary) no real part, must be positive.

[C. XI.

6]

Evidently ϑu is an even function: $\vartheta(-u) = \vartheta u$. Moreover, it is at once seen that we have

$$\vartheta(u + \pi) = \vartheta u, \quad \vartheta(u + i\Pi) = -q^{-1}e^{-iu}\vartheta u,$$

where q is

$$\vartheta(u + \pi + i\Pi).$$

where as u and v are positive or negative integers, the product of ϑu by an exponential factor, or say simply that it is a multiple of ϑu .

Writing $u = \frac{\pi}{2}\xi$, we have $\vartheta v = \frac{1}{2}\sqrt{2K}\vartheta\xi$, that is,

$$\vartheta\left(\frac{\pi}{2}\xi\right) = \xi^{nK},$$

and therefore also

$$\vartheta(\sin \xi + v) = \xi^{nK}\vartheta v.$$

The above properties are general, but if θ be real, then K, K', q being as in Jacobi (consequently q being real, positive, and less than 1, and K and K' real and positive), and assuming $\Omega = \frac{K'}{K}, \pi$, we have in the same thing,

$$q(= e^{-\pi\Omega/\Omega}) = e^{-\pi K'/K},$$

the function ϑv is given (*ut ante*) of Jacobi's ϑv by the equation $\vartheta u = q^{\frac{1}{4}}\vartheta x$.

That is, if Ω is the value of the case $u = v$.

We have, as before, the system of three equations

$$\operatorname{sn} u = \frac{1}{\sqrt{k}} \frac{\vartheta(u, k)}{\vartheta_0(u, k)}, \quad \operatorname{cn} u = \sqrt{\frac{k'}{k}} \frac{\vartheta_0(u, k)}{\vartheta_0(u, k)}, \quad \operatorname{dn} u = \sqrt{k'} \frac{\vartheta_0(u, k)}{\vartheta_0(u, k)}.$$

Consider now the integral

$$\int_a^b \frac{dx}{\sqrt{(x - \alpha, x - \beta, x - \gamma, x - \delta)}} = \int_a^b \frac{dx}{\sqrt{X}} \text{ suppose,}$$

where $\alpha, \beta, \gamma, \delta$ are taken to be real, and the four factors regarded as fixed magnitudes, viz. it is assumed that $b - a$ is the value of the radical at the inferior limit is positive, and as b is between β and δ , X is positive, and *a fortiori* then $\sqrt{X}$ denotes the positive value of the radical; but if α is between β and δ , then X is negative, and we must

that the sign of $\sqrt{X}$ is taken so that $\sqrt{X}$ is equal to a positive multiple of i , and this being so the integral is taken from the inferior limit to the superior limit x , which we may.

Take a linear function of y , such that for

$$x = a, b, c, d,$$

$$y = 0, 1, \frac{1}{k^2}, \infty, \text{ respectively,}$$

so that, x increasing continuously from a to d , y will increase continuously from 0 to ∞ . We have

$$y = \frac{c-a}{c-b} \cdot \frac{x-b}{x-a},$$

$$y = \frac{c-a}{c-d} \cdot \frac{x-d}{x-a},$$

$$1-y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

$$1-k'y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

and, thence,

$$\sqrt{y \cdot 1-y \cdot 1-k'y} = \frac{(a-c)\sqrt{X}}{(c-a)\sqrt{(x-a) \cdot (x-a)}},$$

where $\sqrt{\frac{(c-d)}{(c-d)}}$ is taken to be positive, and the signs of $\sqrt{X}$ is fixed as above. For y between 0 and 1, $y = \frac{1}{x^2}$, $1-y$, $1-k'y$ will be positive, and $\sqrt{y}$ being between 1 and $\frac{1}{x^2}$, $1-y$ will be negative, but $1-k'y$ will be positive, viz. the radical is then taken as a positive multiple of i .

We have moreover

$$dy = \frac{c-a}{c-b} \cdot \frac{dx}{(x-a)^2},$$

and therefore

$$\frac{dy}{\sqrt{y \cdot 1-y \cdot 1-k'y}} = \sqrt{(c-d) \cdot (c-a)} \cdot \frac{dx}{\sqrt{X}},$$

where $\sqrt{(c-d) \cdot (c-a)}$ is positive; or, say,

$$\int \frac{dy}{\sqrt{y \cdot 1 - y \cdot 1 - k'y}} = \sqrt{(c-d) \cdot (c-a)} \cdot \int \frac{dx}{\sqrt{X}}.$$

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Hence, writing $y = \sin^2 u$, we have

$$\Pi u = \sqrt{(c-d) \cdot (c-a)} \int_a^b \frac{dx}{\sqrt{X}},$$

and it is to be further noticed that to

$$x = a, b, c, d,$$

correspond

$$\Pi u = 0, \frac{1}{2}K, \frac{1}{2}K + \frac{1}{2}K'.$$

or we may say

$$u = 0, K, K + iK', 2K + iK'.$$

Writing for shortness

$$\frac{1}{\Pi} = \sqrt{(c-d)(c-a)} = u,$$

we have

$$\sin u = \int_a^b \frac{dx}{\sqrt{X}},$$

and moreover

$$\sin(K + iK') = \int_a^b \frac{dx}{\sqrt{X}},$$

$$n(K + iK') = \int_b^c \frac{dx}{\sqrt{X}},$$

$$n(2K + iK') = \int_c^d \frac{dx}{\sqrt{X}},$$

or if for a moment we write

$$\int_a^b \frac{dx}{\sqrt{X}} = A, \text{ \&c.}$$

then these equations are

$$\begin{aligned} nK &= A, \\ n(K + iK') &= A + B, \\ n(2K + iK') &= A + B + C. \end{aligned}$$

Hence $B + C = 2A = B$, or $A = B - C$, or $B - C + D = 0$, that is,

$$\int_a^b \frac{dx}{\sqrt{X}} = \int_b^c \frac{dx}{\sqrt{X}},$$

where observe as before that $x = a$ to $x = b$, or $x = c$ to $x = d$, X is positive, and the radical $\sqrt{X}$ is taken to be positive.

We have also

$$\begin{aligned} nK &= A = \int_a^b \frac{dx}{\sqrt{X}}, \\ niK' &= C - B = \int_a^b \frac{dx}{\sqrt{X}}. \end{aligned}$$

when, as before, from b to c , X is negative, and the sign of the radical is such that $\sqrt{X}$ is a positive multiple of i ; the last formula may be more conveniently written

$$niK' = \int_b^c \frac{dx}{\sqrt{-X}},$$

where, here b to c , $-X$ is positive, and $\sqrt{(-X)}$ is to be taken as positive.

Collecting the results, we have

$$\int_a^b \frac{dx}{\sqrt{X}} = \frac{nK}{\sqrt{(c-d)(c-a)}}, \quad b \leq a \leq b, \dots b \leq b \leq b \leq a,$$

and also

$$k = \frac{c-b}{a-b} \cdot \frac{d-a}{c-a},$$

and thence conversely

$$k' = \frac{a+d-b-c + (b-c-a+d) \sin^2 n}{a+b-c-d + (b-c+a-d) \sin^2 n},$$

or, what is the same thing,

$$\begin{aligned} nu^2 &= \frac{b-d \cdot x-a}{b-a \cdot x-d}, \\ nu^2 &= \frac{c-d \cdot x-a}{c-a \cdot x-d}, \\ nu^2 &= \frac{d-a \cdot x-b}{a-b \cdot x-d}, \\ nu^2 &= \frac{b-d \cdot x-a}{b-a \cdot x-d}, \end{aligned}$$

where, in place of the elliptic functions nu we substitute their ϑ -values, it will be recollected that Ω , the parameter of the ϑ -functions, has the value

$$\frac{K'}{K}\pi = \left(= \frac{\Omega'}{\Omega} \pi \right) = \pi \frac{\int_a^b \frac{dx}{\sqrt{-X}}}{\int_a^b \frac{dx}{\sqrt{X}}},$$

and, so before,

$$K = \int_a^b \frac{dx}{\sqrt{X}},$$

Hence, finally, a, b, K, K', Ω denoting given functions of a, b, c, d , as above

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

we have severally

$$\begin{aligned} \frac{b-d}{c-d} \cdot \frac{a-c}{c-a} &= \frac{1}{k^2} \frac{\vartheta_1^2 u}{\vartheta_2^2 u}, & \frac{\vartheta_3^2 u}{\vartheta_2^2 u}, \\ \frac{d-a}{d-b} \cdot \frac{b-c}{a-c} &= \frac{k'^2}{k} \frac{\vartheta_2^2 u}{\vartheta_3^2 u}, & \frac{\vartheta_1^2 u}{\vartheta_2^2 u} + \frac{1}{2} K', \\ \frac{c-a}{d-a} \cdot \frac{b-d}{b-c} &= k'^2 \frac{\vartheta_4^2 u}{\vartheta_3^2 u}, \end{aligned}$$

which are the formulae in question.

The problem is to obtain these (and this in the more general case where a, b, c, d have any given (real or imaginary) values directly from the assumed equation

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

and from the foregoing definition of the function ϑu .

It may be recalled that the function ϑu is a doubly infinite product

$$\vartheta u = \prod \prod \left[1 + \frac{u}{m\pi + (n + \frac{1}{2})i\Pi} \right]$$

as and a positive or negative integers from $-\infty$ to $+\infty$; I purposely omit all further explanations as to limits, or what is the same thing,

$$\vartheta \frac{\pi}{2K} = \prod \prod \left[1 + \frac{u}{2mK + (2n + 1)iK'} \right]$$

and consequently that, disregarding constant and exponential factors, the foregoing expressions of

$$\frac{b-d}{c-d} \cdot \frac{a-c}{c-a}, \quad \frac{d-a}{d-b} \cdot \frac{b-c}{a-c}, \quad \frac{c-a}{d-a} \cdot \frac{b-d}{b-c},$$

are the squares of the expressions

$$\frac{X}{Y}, \frac{Y}{Z}, \frac{Z}{W}, \quad \text{where } X, Y, Z, W \text{ are respectively of the form}$$

$$\begin{aligned} & n \prod \prod \left[1 + \frac{u}{(2n+1)iK'} \right], \quad \prod \prod \left[1 + \frac{u}{(2m+1)iK'} \right], \\ & \prod \prod \left[1 + \frac{u}{2mK + iK'} \right], \quad \prod \prod \left[1 + \frac{u}{(2m+1)K + iK'} \right], \end{aligned}$$

where $(m, n = 2mK + 2niK', \&c.,$ the *smha* over the m or the n denotes that the $2m$ or the $2n$ as the case may be) is to be changed into $2m+1$ or $2n+1$. But this is a transcription which has apparently no application to the ϑ -functions of more than one variable.

717.

ON THE TRIPLE THETA-FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 48—50.]

AS a specimen of mathematical notation, viz. of the notation which appears to me the easiest to read, and also to print, I give the definition and demonstration of the fundamental properties of the triple theta-functions.

Definition.

$$\vartheta(U, V, W) = \Sigma \exp \Theta,$$

where

$$\Theta = (A, B, C, F, G, H)(l, m, n)^2 + (U, V, W)(l, m, n).$$

l , denoting the sum in regard to all positive and negative integer values from $-\infty$ to $+\infty$ (zero included) of l, m, n respectively.

$\vartheta(U, V, W)$ is considered as a function of the arguments (U, V, W) , and it depends also on the parameters (A, B, C, F, G, H) .

First Property. $\vartheta(U, V, W) = 0$, for

$$U = \frac{1}{2}\pi i + (A, H, G)(\alpha, \beta, \gamma),$$

$$V = \frac{1}{2}\pi i + (H, B, F)(\alpha, \beta, \gamma),$$

$$W = \frac{1}{2}\pi i + (G, F, C)(\alpha, \beta, \gamma),$$

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta$ being any positive or negative integer numbers, such that as $\alpha + \beta + \gamma = \text{odd number}$.

Demonstration. It is only necessary to show that to each term of ϑ there corresponds a second term, such that the inclusion of the two exponentials differ by an odd multiple of πi .

Taking l, m, n the integers which belong to the one term, those belonging to the other term are

$$-(l + a), -(m + \beta), -(n + \gamma).$$

(where observe that one at least of the numbers α, β, γ being odd, this system of values is not in any case identical with l, m, n). The two exponents are

$$\Theta_i = (A, B, C, F, G, H)(l, m, n)^2 + (U, V, W)(l, m, n),$$

and

$$\Theta'_i = (A, B, C, F, G, H)(l+a, m+\beta, n+\gamma)^2 - (U, V, W)(l+\alpha, m+\beta, n+\gamma),$$

viz. the value of Θ' is

$$\begin{aligned} &= (A, B, C, F, G, H)(l, m, n)^2 + 2(A, H, G)(l\alpha, \beta, \gamma) + 2(H, B, F)(m\alpha, \beta, \\ &+ (A, B, C, F, G, H)(a, \beta, \gamma)^2 - (U, V, W)(l, m, n) - (U, V, W)(\alpha, \beta, \gamma), \end{aligned}$$

and we then have

$$\begin{aligned} \Theta' - \Theta &= -2(A, B, C, F, G, H)(l, m, n)^2 + 2(A, H, G)(\alpha, \beta, \gamma)l + 2(H, \\ &+ 2(G, F, C)(\alpha, \beta, \gamma)n + (A, B, C, F, G, H)(a, \beta, \gamma)^2 \\ &- 2(U, V, W)(l, m, n) - (U, V, W)(\alpha, \beta, \gamma). \end{aligned}$$

which proves the theorem.

As to the notation, remark that, after (A, B, C, F, G, H) has been written out in full, we may instead of $((A, B, C, F, G, H)(\alpha, \beta, \gamma)^2$ write

$$(\alpha, \beta, \gamma)^2,$$

and that we may use the like abbreviation

$$(A, \dots)(\alpha, \beta, \gamma) \text{ for } (A, H, G)(\alpha, \beta, \gamma), \text{ \&c.},$$

$$(H, \dots)(\alpha, \beta, \gamma) \text{ for } (H, B, F)(\alpha, \beta, \gamma), \text{ \&c.},$$

$$(G, \dots)(\alpha, \beta, \gamma) \text{ for } (G, F, C)(\alpha, \beta, \gamma), \text{ \&c.}$$

These are not only abbreviations, but they make the formula actually clearer, as bringing them into a similar compact; and I accordingly use them in the demonstration which follows.

Second Property. If U, V, W , denote

$$\begin{aligned} U' &= \pi i + (A, H, G)(\alpha, \beta, \gamma), \\ V' &= \pi i + (H, B, F)(\alpha, \beta, \gamma), \\ W' &= \pi i + (G, F, C)(\alpha, \beta, \gamma), \end{aligned}$$

respectively, where $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$ are any positive or negative integers (zero values admissible), then

$$\vartheta(U+u, V+v, W+w) = \exp \left[-2(\alpha U + \beta V + \gamma W) - (\alpha, \beta, \gamma)^2 \right] \vartheta(U, V, W),$$

or, say

$$= \exp \left[-(A, \dots)(\alpha, \beta, \gamma)^2 - 2(U, V, W)(\alpha, \beta, \gamma) - (\alpha, \beta, \gamma)^2 \right] \vartheta(U, V, W)$$

Demonstration. Writing $\vartheta(U, V, W) = \Sigma \exp \Theta$, then in the expression of Θ , we may in place of l, m, n write $-l - \alpha, -m - \beta, -n - \gamma$; we thus obtain

$$\begin{aligned} \Theta_l &= (A, \dots)(l, m, n)^2 + 2(A, \dots)(l, m, n)(\alpha, \beta, \gamma) + (A, \dots)(\alpha, \beta, \gamma)^2 \\ &\quad + (U, V, W)(l, m, n) + (U, V, W)(\alpha, \beta, \gamma), \end{aligned}$$

which is

$$\begin{aligned} &= (A, \dots)(l, m, n)^2 \\ &\quad + 2(\alpha, \beta, \gamma)(\alpha U + \beta V + \gamma W) - 2(A, \dots)(\alpha, \beta, \gamma) + \Sigma'(A, \dots)(\alpha, \beta, \gamma)^2 - \Sigma a^2(A, \dots) \\ &\quad - 2(\alpha, \beta, \gamma)^2 + 2\alpha, \beta, \gamma) + (A, \dots)(\alpha, \beta, \gamma)^2 - 2(A, \dots)(\alpha, \beta, \gamma) \\ &\quad + (U, V, W)(l, m, n) + (U, V, W)(\alpha, \beta, \gamma), \end{aligned}$$

which is

$$\begin{aligned} &= (A, \dots)(l, m, n)^2 \\ &\quad + \Sigma'(A, \dots)(\alpha, \beta, \gamma)^2 + \Sigma a^2(A, \dots)(\alpha, \beta, \gamma) + \Sigma \alpha \beta (A, \dots)(\alpha, \beta, \gamma)^2 \\ &\quad + (U, V, W)(l, m, n) + \Sigma(A, \dots)(\alpha, \beta, \gamma) \end{aligned}$$

Hence, rejecting the last line, which (as on can readily verify) has the exponential multiplied as the case may be, $\vartheta(U, V, W)$ as $\vartheta(U, V, W)$ multiplied by the factor

$$\exp \left[-(A, \dots)(\alpha, \beta, \gamma)^2 - 2(U, V, W)(\alpha, \beta, \gamma) - (\alpha, \beta, \gamma)^2 \right],$$

which is the theorem in question.

In many cases a formula, which belongs to an indefinite number of k letters, is most easily intelligible when written out for three letters, but it is sometimes convenient to speak of the letters l , m , ... or even the letter l , and to write out the formula accordingly.

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In fact, writing herein $\alpha = \gamma = 0$, that is, $a = \gamma = 0$, the right-hand side becomes a^2 and the arts on the left-hand side are $a = B$, $A = B$, $a = A = I$, which suppose any four sum the sum of which is $= 0$.

Writing in the last-mentioned equation α, β, γ for the a 's of A, B, γ, δ respectively, the equation becomes

$$\begin{aligned} P &= \beta P, \\ Q &= 1 - a^2 + \beta^2 + By^2, \\ R &= 1 - By^2 - 2\alpha + By^2, \\ S &= 1 - By^2. \end{aligned}$$

L, M, N as the like functions $\alpha, \beta, \gamma, \delta$ respectively, A, B, C, D being the same as before. And in fact

$$\begin{aligned} -LQ &= -y^2(\alpha^2 + \alpha^2) + \beta^2 + (\beta^2 - \alpha^2) + B(y^2 - x^2) + B(\beta - \alpha) + \dots, \\ &= \frac{1}{B} [\beta^2 - (\beta - \alpha)^2 + By^2 - By^2] + B(y^2 - x^2) + \dots, \\ &= \frac{1}{B} [\beta^2(x^2 - y^2) + B(y^2 - x^2) + B^2(y^2 - x^2)], \end{aligned}$$

that is,

$$\begin{aligned} -LPQ &= -y^2(\beta^2 - \alpha^2) + \beta^2 - (\beta - \alpha) + \dots, \\ &+ (B + \alpha - \beta)\beta(y^2 - x^2) + \beta - B^2(y^2 - x^2) + \dots, \end{aligned}$$

or omitting the factor β , that is,

$$-LPQ = -y^2(\beta^2 - \alpha^2) + \beta^2 - y^2(x^2 - \alpha^2) + \dots + \beta(y^2 - x^2) - By^2(x^2 - \alpha^2),$$

It is easy to verify the form of the orders 0, 1, 2, 3, and 4 in x, y, z, w separately destroy each other, for instance, for the terms of the order 4, the value of Σ is then

$$\begin{aligned} &= A^2 + \Sigma y^2(\beta^2 - \alpha^2) + y(x^2 - \alpha^2) + B^2(x^2 - \alpha^2) + (\Sigma \beta^2)(x^2 - \alpha^2) + \dots, \\ &\frac{1}{B} [\Sigma \beta a^2(x^2 + y^2) + \Sigma a^2 B^2(x^2 - y^2) + B^2 a^2(x^2 - y^2)], \end{aligned}$$

It is easy to verify that the terms of the orders 0, 1, 2, 3, and 4 in x, y, z, w separately destroy any sum sum of x which is

$$-LP^2 + yQ^2 + yP^2 + xR^2 + yS^2 + (LP + yQ + xR + yS + yT)$$

$+(D^2-1-LP^2)(yP^2+yP^2-yQ^2-yQ^2)+2(LP+yQ+xR+yS+yT)(LP+yQ+x$
or, omitting the factor β , that is,

$$-(LP+yQ^2+xR+yS+yT)+\Sigma(LP+yQ+xR+yS+yT)(LP+yQ+xR+yS+yT)$$

The theorem in the original form was obtained by me as follows:
using the elliptic coordinates p, q, r , such that

$$\frac{p^2}{a+p} + \frac{q^2}{b+p} + \frac{r^2}{c+p} = 1,$$

or, what is the same thing,

$$-Bp\rho^2 = a + p \cdot b + p \cdot c + p,$$

$$-y\eta y = b + q \cdot a + q \cdot c + q,$$

$$-z\zeta z = c + r \cdot a + r \cdot b + r,$$

where A, B denote $b - c, c - a, a - b$ respectively; then, treating r as a constant, the coordinates x, y, z will belong to a point on the ellipsoid

$$\frac{x^2}{a+r} + \frac{y^2}{b+r} + \frac{z^2}{c+r} = 1,$$

and the differential equation of the right lines upon this surface is

$$\frac{dp}{\sqrt{a+p \cdot b+p \cdot c+p}} + \frac{dq}{\sqrt{a+q \cdot b+q \cdot c+q}} = 0.$$

Take a, b, c , the coordinates of a point on the surface, and p, q , the corresponding values of p, q , so that

$$-Bp_0\rho_0^2 = a + p_0 \cdot b + p_0 \cdot c + p_0,$$

$$-Aq_0\eta_0 = b + q_0 \cdot a + q_0 \cdot c + q_0,$$

$$-z\zeta_0 = c + r \cdot a + r \cdot b + r,$$

then the equation of the tangent plane at the point (a, b, c, p_0, q_0) is

$$\frac{xz_0}{a+r} + \frac{yy_0}{b+r} + \frac{zz_0}{c+r} = 1,$$

or, substituting for x, y_0, z_0 their values, we have

$$= \beta p_0 \cdot ay + \cdot bxz + Az + \dots,$$

and consequently the equation of the tangent plane through (a, b, c, p_0, q_0) is the integral of the proposed differential equation.

Writing

$$mz^2 = A(a+p), \quad mz^2 = B(b+p), \quad dx^2 = C(c+p),$$

the values of A, B, C, S are determined; and, assuming for q, r, p , the like forms with the arguments a, a_0, a_0 , the differential equation becomes $dz + dz$, having the

$$[\quad \quad \quad 10 - 2]$$

integral $u + x_0 + v = u_0$; while the foregoing integral equation, on reducing the constant coefficients contained therein, takes the form

$$\begin{aligned}
 -l^2xx + mz_0mz + nz_0mz_0 + nz_0nz + nz_0nz_0 + \dots &= 0, \\
 &= \frac{1}{2}lm \cdot mz_0mz_0, \\
 &= -l^2nz \cdot mz_0mz_0, \\
 &= \frac{nz_0}{B^2 - 1 - l^2} \cdot \frac{l^2}{l^2 - l^2},
 \end{aligned}$$

via. this equation holds good if $a = u_0 + v$, we have the theorem.

If, as above, $a = u + v + v$, $b = u - v + v$, $c = u + v - v$, where r is regarded as constant, then the three products $\text{sn}u \text{nz} \text{nz}$, $\text{cn}u \text{nz} \text{nz}$, $\text{dn}u \text{nz} \text{nz}$, and equally so the three products $\text{sn}z \text{nz} \text{nz}$, $\text{cn}z \text{nz} \text{nz}$, $\text{dn}z \text{nz} \text{nz}$, are expressed by means of those products with the u , v , w of the u , of the v , viz. for two surfaces it would be possible to obtain a real relation equation, which may be expressed in the squares of the u , v , w of the r , of the z , all of whose internal coefficients are nominal, viz. the such would be obtained, in fact, a similar relation in the like form by giving to the parameter r the advantage of being of a degree which is the half of the equation which involves only the squared functions.

Write a , b , c or for $\text{sn}u$, $\text{cn}u$, $\text{dn}u$, respectively; then, writing

$$\begin{aligned}
 A &= \sqrt{1 - y^2 \cdot 1 - l^2y^2}, & a &= \sqrt{1 - l^2 \cdot 1 - l^2x^2}, \\
 B &= \sqrt{1 - l^2 \cdot 1 - l^2y^2}, & b &= \sqrt{1 - x^2 \cdot 1 - l^2y^2}, \\
 C &= \sqrt{1 - l^2 \cdot 1 - l^2y^2}, & c &= \sqrt{1 - l^2 \cdot 1 - x^2 - y^2}, \\
 D &= 1 - l^2x^2y^2, & d &= 1 - l^2z^2x^2,
 \end{aligned}$$

we have

$$\text{sn}(z + v) = \frac{xA + ya}{D},$$

that is,

$$\frac{A + P}{D} = \frac{\sigma c}{\sigma a \cdot \sigma b},$$

and consequently

$$Dn = (A - x)(A + x),$$

$$PD = (P + x)(P - x);$$

whence

$$Dn = PD = (Aa - x)(Aa - x).$$